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Modelling denitrification process in a static mixer–reactor using lattice-Boltzmann method

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ABSTRACT

Nitrate removal in wastewater systems is crucially important to maintaining the quality of potable water as well as protecting vulnerable ecosystems. In this study, a denitrification model was incorporated into lattice-Boltzmann model (LBM) to investigate denitrification efficacy of wastewater treatment in a static mixer–reactor. The model was used for simulating the effects of six solid and porous obstacle configurations, different inlet velocities, and inlet methanol concentrations on denitrification within a static mixer–reactor with six different configurations. The results show that all the obstacles and porous blocks could improve the denitrification efficiency. However, the performances of these configurations could be affected differently by the inlet conditions. At a low velocity ($u = 0.5u_0$), the denitrification rates are almost twice as that at $u = 1.5u_0$. However, at a high inlet velocity, there were small differences of the denitrification efficiency among different configurations. Furthermore, the rectangular obstacle had the greatest effects on the denitrification rates at the low inlet velocity while its performance at a higher velocity was almost comparable to those of the porous blocks. It is also found that at a high inlet methanol concentration ($C_{2\text{-u}} = 2.0C_{2\text{-u0}}$), all the configurations increased the denitrification efficiency of up to 10%, compared to clear channel. However, the rectangular obstacle resulted in a decrease of about 3% in denitrification at a low inlet methanol concentration ($C_{2\text{-u}} = 0.5C_{2\text{-u0}}$) while all other configurations could improve the performance. Although at a low methanol concentration, the solid obstacles could dwindle the performance, the porous blocks did enhance denitrification rates in all the inlet methanol concentrations. This implies that the porous configurations have a wider range of operational conditions, compared to other obstacle configurations. A proper design of static mixer–reactors without any moving part is able to improve the denitrification efficiency in wastewater treatment. The present model is a useful tool to evaluate the performance of different reactor configurations under combinations of all significant parameters, such as inlet velocity and methanol concentrations.

1. Introduction

Ammonia (NH_3) and nitrates (NO_3^-) are common components of ammonia-based fertilizers, and wastewater. The nitrates originate mainly from nitrification reactions via chemo-lithotrophic bacterial oxidation of ammonia to nitrate (Rahimi et al., 2011; Norton et al., 2002). Concentrations of nitrates are globally increasing in lake and river systems due to runoff from increased fertilizer applications and sewage discharge. Increased nitrate concentration in ponds and lakes also results in eutrophication and algal blooms, which can damage water quality and the viability of an ecosystem (Ghafari et al., 2008). Nitrates are also a contaminant in water that can cause a variety of health issues, such as infantile methemoglobinemia (blue baby syndrome) (Wang and Chu, 2016) and is reduced to carcinogens, such as nitriles in saliva, (Matějů et al., 1992). Nitrate removal in water systems is crucially important to maintaining the quality of potable water as well as protecting vulnerable ecosystems.

A variety of technologies have been developed for the purpose of nitrate removal from wastewater, such as ion exchange, adsorption, membrane separation, electrodialysis, chemical denitrification, and

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biological denitrification (Aslan and Türkman, 2004; Jianlong and Jing, 2005). Biological denitrification is one of the key processes in the purification of wastewater where nitrate is removed from water in anaerobic environments. In biological denitrification, autotrophic microbes utilize organic carbon as a carbon source to convert NO_3^- to NO_2^-, NO, and N_2O in the sequence reductive reaction and eventually to N_2O or N_2 gas, depending on oxygen availability (Van Rijn et al., 2006; Karanasios et al., 2010). Denitrification is irreversible once NO is formed, and gas released is taken out of the system, resulting in the net release of N_2 or N_2O gas, depending on the extent of anaerobicity in the environment (Martens, 2005).

A number of studies have been performed in different reactors (Tong et al., 2014) on the different denitrification pathways, such as the anaerobic ammonia oxidation (anammox) pathway (Kartal et al., 2011), the HDN (heterotrophic denitrification) pathway (Pan et al., 2013) and the ammonia oxidizing bacteria (AOB) pathway (Schreiber et al., 2012) in the past years. Xing et al. (2020) investigated a novel double-layered microsphere design for the purpose of regulating carbon release speed to limit excessive concentrations of dissolved organic carbon (DOC). Much research has been conducted on testing the efficacy of different carbon sources such as methanol (Dold et al., 2008; Martineau et al., 2013; Pan et al., 2012), wood chips (Hu et al., 2017) or corn husks (He et al., 2016) in wastewater treatment plants. However, it has been found that denitrifying processes could release large amounts of N_2O into the atmosphere, a greenhouse gas with approximately 300 times the long-term effect of CO_2, under some conditions (Li et al., 2020; Chen et al., 2020). Some studies have been conducted to improve the rate of complete denitrification to releasing N_2 rather than N_2O. Mannina et al. (2018a, b) investigated the correlation between biofilm growth and N_2O release. They found that the presence of biofilm reduced emissions of N_2O in the membrane reactor. Biofilm reactors have higher denitrification rates than fixed-bed or fluidized bed reactors because biofilm carriers have high surface area to increase biomass (Vasiliadou et al., 2009; Rezania et al., 2005; Ghafari et al., 2009). Stein (2019) showed that the different pathway of denitrification can reduce nitrate to either N_2O or N_2. A strain of denitrifying bacteria (NNA5) has also been identified to be highly efficient in reducing N_2O to N_2, to the extent, leading to zero N_2O emissions when the substrate is nitrate or nitrite (Liu et al., 2016). Wang and Chu (2016) evaluated effects of different parameters on denitrification such as the microbial community of biofilm, the varieties of solid carbon source, and the cost of the process. Cerminara et al. (2020) studied the denitrification capacity of old municipal solid waste (MSW) in six landfill bioreactors with very low COD/NO_3^- N mass ratios. They found that denitrification occurred in all bioreactors even at substantially low concentrations of biodegradable organic matter.

The efficiency and rate of mixing in reactors play an important role because N_2O or N_2 production is depending on mixing of nitrate, organic carbon, and oxygen. There are two common types of mixers: dynamic and static mixers. Dynamic mixers (agitated tanks) are used for batch operations while static mixers, as inline mixing devices, can be used in continuous operations or recycle loops. Therefore, the static mixed reactor can be used as a replacement of conventional agitators as they can have similar or higher performance at lower cost (Myers et al., 1997; Thakur et al., 2003; Delavar, 2012; Ajarostaghi et al., 2020). T-micro mixers are a relatively simple static mixer that consists of two different inlets and a mixing channel. As a type of static mixer–reactor, they have many advantages, such as low energy consumption, low or no maintenance cost, a short residence time, and small space requirements due to no moving parts. These advantages have made it practical for use in some important engineering applications such as combustion, biomedical devices, pharmaceutical crystallization, catalytic reaction, dust collection, liquid–liquid extraction, absorption, ventilation systems, etc. Static mixer–reactors are an important component within denitrification plants (Lang et al., 1995; Ghanem et al., 2014; Zidouni et al., 2015). Because turbulent diffusion may be low in distributing reaction components, many static mixers employ fluid physics systems to increase the homogeneity of a fluid and in turn the reaction rate. In the past decades, research has been done in the field of miniaturizing mixers to a micro scale (Nguyen and Wu, 2005), which have excellent heat and mass transfer performances due to an extremely large surface area to volume ratio (Aru et al., 2016). Such mixers can also be used to remove NO gas from effluent. The inclusion of the geometric configuration of a static mixer in a wastewater treatment plant could positively enhance the denitrification rate and reduce NO emissions. Although configurations of a static mixer–reactor can significantly affect the efficiency of denitrification efficiency in wastewater treatment due to poor mixing conditions, there is a major challenge to compare experimentally different configurations designs due to a wide range combination of obstacles, channels and operational parameters. Therefore, it is still unclear how reactor configurations enhance mixing and denitrification efficiency in the wastewater treatment reactors.

A computer modelling of denitrification would be a powerful asset as it would save time and cost in reactor design and development. In the past years, many models of denitrification processes in wastewater treatment have been developed using computational fluid dynamics (CFD) (Climent et al., 2018; Vilardi et al., 2020), reaction diffusion equations (Vasiliadou et al., 2009) or kinetic models (Gao et al., 2010). However, these existing models have weakness when resolving non-equilibrium dynamics on the interface, such as porous media, multiphase flow and multi-component flow (Alazmi and Vafai, 2001; Ajarostaghi et al., 2020). Compared to the conventional CFD approaches, lattice Boltzmann models (LBM), as a mesoscopic method, has advantages in simulating complex, non-equilibrium phenomenon and processes in microscale (Delavar, 2012; Delavar and Hedayatpour, 2012; Ajarostaghi et al., 2020; Delavar et al., 2010; Sipey et al., 2019; Tian and Wang, 2018, 2019; Delavar and Wang, 2020a, b). This makes it more suitable for handling of complex geometries and multi-component flows in wastewater treatment. Therefore, there is a need to develop modelling capacity of LBM in evaluating effects of reactor structures and carbon source on the denitrification processes.

The objectives of this study are to simulate the mixing and denitrification processes in different configurations of a static mixer–reactor using LBM. We will incorporate a module of denitrification processes into LBM framework, and then, apply it to evaluate effects of mixing and configurations on denitrification processes for the performance improvement of the static mixer–reactor in the wastewater treatment.

2. Material and methods

2.1. *Denitrification processes*

Nitrate and carbon sources are mostly from the wastewater influent. Additional external organic carbon, such as methanol or acetate, is often applied if organic carbon is insufficient in wastewater for removing nitrate. Heterotrophic denitrification includes sequential redox reactions from NO_3^- to NO_2^-, NO, N_2O and finally to N_2 using organic carbon as the electron donor. Stoichiometry and kinetics of heterotrophic denitrification in the presence of methanol as carbon source is expressed as:

$$6NO_3^- + 5CH_3OH \rightarrow 3N_2 + 5CO_2 + 7H_2O + 6OH \quad (1)$$

2.2. *Flow equations*

In this study the mixture of water and nutrients passes through the domain; to capture the local fluid velocity (**V**) field, the governing equations, continuity, and the Navier–Stokes equations, should be solved:

$$\nabla.V = 0 \quad (2)$$

$$\rho\frac{\partial V}{\partial t} + \rho\,(T)\,V.\nabla V = -\nabla p + \mu\,(T)\,\nabla^2 V \quad (3)$$

where ρ, P and ϑ denote the density, pressure, and kinematic viscosity, respectively.

2.3. Nutrient transport

In the numerical simulation of denitrification, the species conservation equation for each nutrient in Eq. (4) should be solved to calculate denitrification rate and local nutrient concentration (C_l) of each species (l) in Eq. (1) as:

$$\rho \frac{\partial C_l}{\partial t} + \rho V.\nabla C_l = \rho D_l \nabla^2 C_l + S_l \tag{4}$$

where S_l is the mass generation source (source term) due to reactions for species l, D_l is the effective diffusion coefficient of the species l, and l is species number that is equal to 1 for NO_3^-, 2 for CH_3OH, 3 for N_2, and 4 for CO_2. The source term for species on the left hand side of Eq. (1) are negative as they are reagents that are consumed during the reaction, while it is positive for species in the right hand side of Eq. (1), the products of the reaction. The relations among the source terms of species are distinguished using Eq. (1). For example, the molar source term of NO_3^- is as two times as that of N_2. As denitrification can be treated as a first order chemical reaction for simplicity (Korom, 2007), the source terms for species can be calculated using Eq. (5) (Etheridge et al., 2014):

$$S_l = k \times n_l \times C_1 \times C_2 \tag{5}$$

where k is 0.198 obtained from Zhou and He (2014), n is the ratio of the species involved in the reaction (n_l = 6 for NO_3^-, 5 for CH_3OH and CO_2, and 3 for N_2), C_1 is the local concentration of nitrate, and C_2 is the local concentration of methanol.

Mass diffusion coefficient is calculated at 15 °C can be scaled based on the values at 25 °C using a function as: (Mostinsky, 1996):

$$D = D_{298}[1 + \alpha(T\text{-}298)] \tag{6}$$

where D_{298} is the diffusion coefficient at room temperature (298 K), α = 20× ${\eta_{298}}^{1/2}$ ×$\rho^{-1/3}$, η_{298} is the solvent viscosity at 25 °C, ρ is the solvent density, and T is the new temperature.

3. The lattice Boltzmann model

In this study, the lattice Boltzmann method (LBM) is used for numerical simulations of velocity and concentration fields. The lattice Boltzmann equation based on BGK (Bhatnagar–Gross–Krook) approximation, without external force, for D2Q9 model, to capture flow fields can be written as:

$$f_k(x + c_k \Delta t, t + \Delta t) = f_k(x, t) + \frac{\Delta t}{\tau}\left[f_k^{eq}(x, t) - f_k(x, t)\right] \tag{7}$$

$$f_k^{eq}(x, t) = \omega_k \rho \left[1 + \frac{c_k.U}{c_s^2} + 0.5\frac{(c_k.U)^2}{\varepsilon c_s^4} - 0.5\frac{U.U}{\varepsilon c_s^2}\right]$$

$$\omega_k = (4/9, 1/9, 1/9, 1/9, 1/9, 1/36, 1/36, 1/36, 1/36)$$

$$c_k = (0, 0; 1, 0; 0, 1; -1, 0; 0, -1; 1, 1; -1, 1; -1, -1; 1, -1)$$

where f_k and f_k^{eq} are distribution and equilibrium distribution functions, respectively. ω_k is weighting factor, and c_k is velocity in lattice directions. D2Q9 means two-dimensional lattice Boltzmann model with 9 streaming directions.

In this study, the distribution function g_l is implemented to capture the lth species concentration field. As concentration is a scalar parameter, the corresponding equilibrium distri-

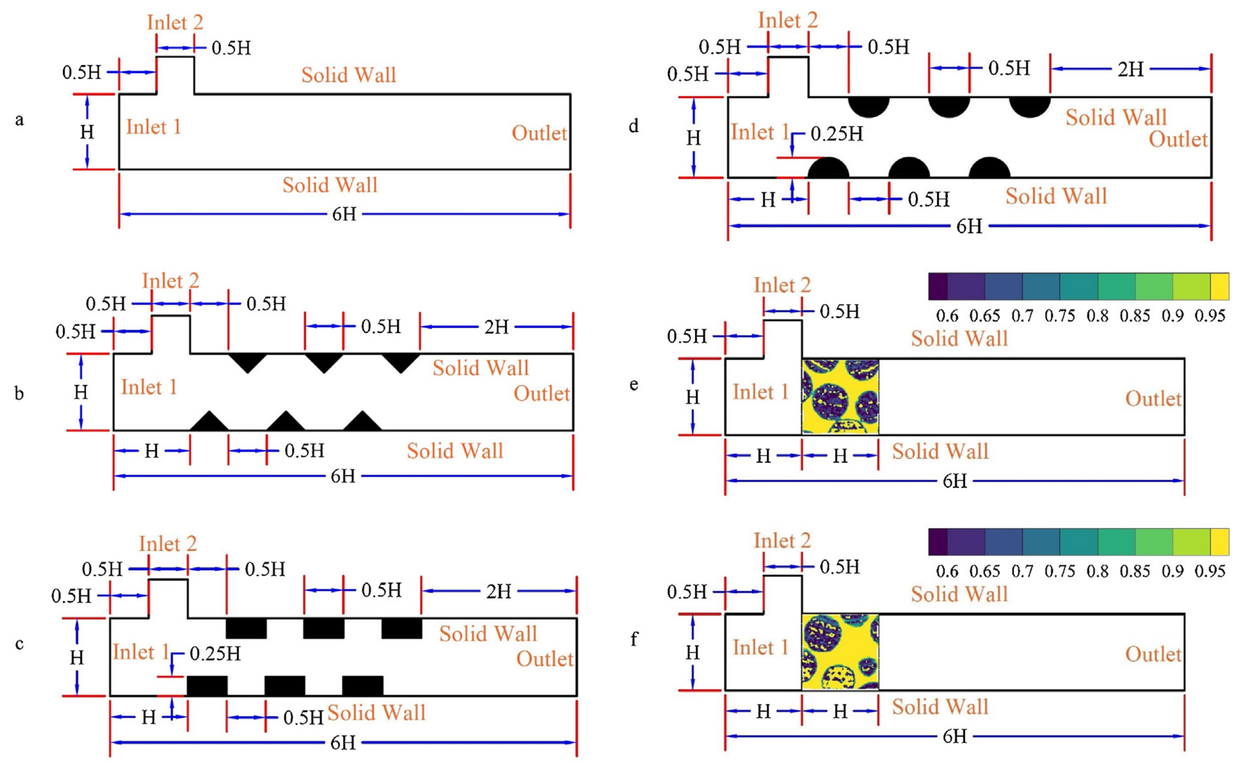

Fig. 1 – T-micro mixer geometries: (a) base geometry (without obstacles), (b) triangular obstacles, (c) rectangular obstacles, (d) circular obstacles, (e) porous block 1 with a porosity of 0.816, and (f) porous block 2 with a porosity of 0.855. The scale on the porous block geometries represents the permeability.

Table 1 – Simulation parameters.

Parameter	Description	Value
–	Species #1	NO_3^-
–	Species #2	CH_3OH
–	Species #3	N_2
–	Species #4	CO_2
–	Species #5	H_2O
–	Species #6	OH^-
u_{inlet}	Average velocity in the inlet	Left inlet: $u = 0.5u_0$, $1.0u_0$, $1.5u_0$ Upper inlet: u
H	Channel height	H = 0.3m
D_l	Mass diffusion coefficient for species (m^2/s) at 25 °C	$D_1 = 1.7 \times 10^{-9}$ (Kreft et al., 2001) $D_2 = 1.563 \times 10^{-9}$ (Derlacki et al., 1985) $D_3 = 2.212 \times 10^{-9}$ (Winn, 1950) $D_4 = 1.6 \times 10^{-8}$ (Berg, 1993) $D_5 = 2.299 \times 10^{-9}$ (Holz et al., 2000) $D_6 = 5.270 \times 10^{-9}$ (Vanysek, 2000)
C_{l-u}	Concentration of species l in upper inlet	$C_{1\text{-}u} = 0$, $C_{2\text{-}u} = 100$, $C_{3\text{-}u} = 0$, $C_{4\text{-}u} = 0$, $C_{5\text{-}u} = 0$, $C_{6\text{-}u} = 0$
C_{l-l}	Concentration of species l in left inlet (mg-N/L)	$C_{1\text{-}l} = 100$, $C_{2\text{-}l} = 0$, $C_{3\text{-}l} = 0$, $C_{4\text{-}l} = 0$, $C_{5\text{-}l} = 0$, $C_{6\text{-}l} = 0$
S_l	Mass source term	$S_1 = -0.198 \times 6 \times C_1 \times C_2$ $S_2 = -0.198 \times 5 \times C_1 \times C_2$ $S_3 = 0.198 \times 3 \times C_1 \times C_2$ $S_4 = 0.198 \times 5 \times C_1 \times C_2$
ε	Porosity	$\varepsilon_1 = 0.816$ $\varepsilon_2 = 0.855$

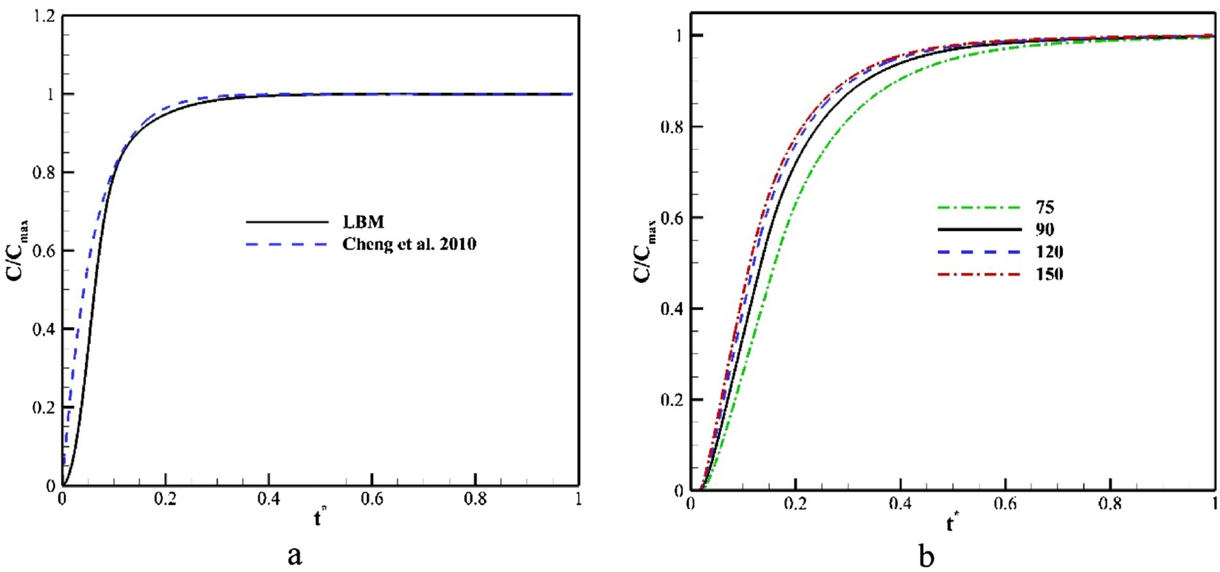

Fig. 2 – (A) Comparison of numerical simulation to literature results for the concentration of CO_2 in the reactor, and (b) normalized carbon dioxide concentration over time for different grid numbers used for the domain's height (1.5 H).

bution functions were defined as below (Delavar and Wang, 2020a, b, 2021):

$$g_{lk}(x + c_k\Delta t, t + \Delta t) = g_{lk}(x, t) + \frac{\Delta t}{\tau_l}\left[g_{lk}^{eq}(x, t) - g_{lk}(x, t)\right] + \Delta t\omega_k S_l \quad (8)$$

$$g_{lk}^{eq}(x, t) = \omega_k C_l\left[1 + \frac{c_k.U}{c_s^2}\right]$$

where S_l is the source term of the lth species as Eq. (5). The macro parameters can be calculated using the distribution functions:

$$\rho = \sum_k f_k,\ \rho u_i = \sum_k f_k c_{ki},\ \text{and}\ C_l = \sum_k g_{lk} \quad (9)$$

where subscript, i, denotes the component of the Cartesian coordinates.

4. Computational domain and boundary conditions

4.1. Mixe–reactor domain and boundary conditions

The mixing reactor can be considered as two-dimensional (2D) because of its symmetry (Fig. 1). Three shapes of obstacles (i.e., triangular, rectangular, and circular), and two different porous blocks are inserted in a static mixer–reactor. These obstacles are used for a comparison of enhanced mixing with clear mixer (without obstacles). Two porous blocks have almost similar porosities of 0.816 and 0.855, respectively but they have different tortuosity structures, as shown in Fig. 1e & f. The left inlet is for nitrate ions, and the upper inlet is for methanol,

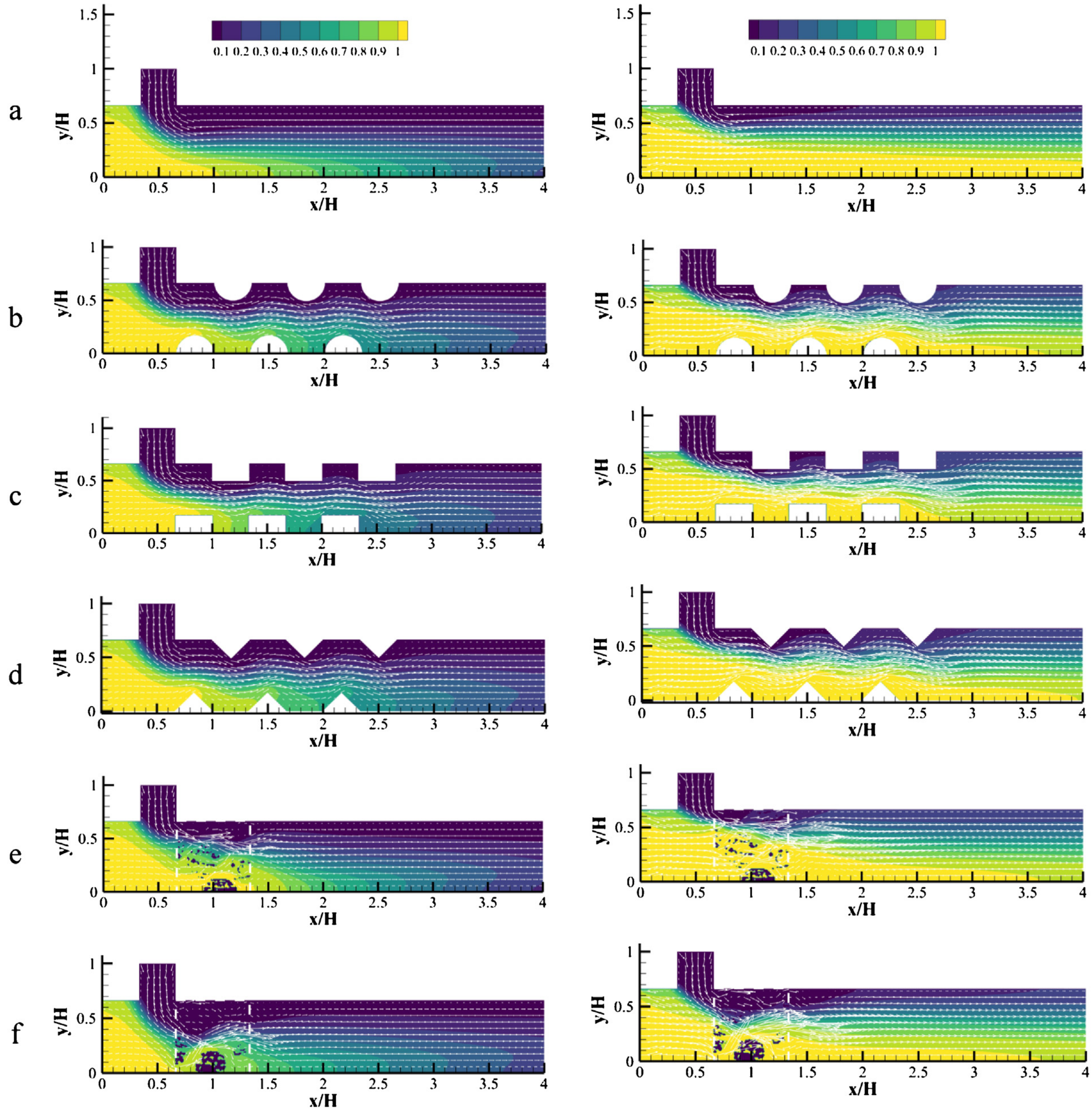

Fig. 3 – Normalized nitrate ion concentration in the domains with (a) no obstacles, (b) circular obstacles, (c) rectangular obstacles, (d) triangular obstacles, (e) porous block configuration 1, and (f) porous block configuration 2 for different inlet velocities of $u = 0.5u_0$ and $u = 1.5u_0$ (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.).

and the right boundary is considered as outlet. All solid walls, including obstacles, are set as bounce back boundaries. Properties, such as viscosity and mass diffusion coefficients, are assumed fixed. The simulations are conducted with fixed inlet velocities and concentrations within each iteration in an isothermal and adiabatic environment to minimize factors that may influence the modelling. H was settled to 30 cm. The 2D simulations were performed after considering symmetry to reduce computational time with a sufficient resolution. Simulation parameters are provided in Table 1.

4.2. *Validation and calibration*

The analytical solution of the concentration of NO_3^- presented by Cheng et al. (2010) as ($C = C_0e^{-kt}$) is used for the LBM model validation when $k = 0.198$ g-N/(m^2 d) at 15 °C (Zhou and He, 2014). Fig. 2a shows a comparison between the analytical and numerical results. It can be seen that the simulated results agree well with the analytical ones. Different grid numbers, 75, 90, 120 and 150 were applied for the domain's height (1.5 H) in Fig. 1 to find the grid-independent solution. Fig. 2b shows the results of different grid sizes. It can be found in Fig. 2b that there is apparent difference between grid 75, 90 and 120 but the difference between grid 120 and 150 is very small. Therefore, the grid number of 120 is used for all simulations which can achieve the accuracy of the results under reasonable CPU time.

5. Results and discussion

In this study, the lattice Boltzmann method is used to investigate mixing and denitrification in six different configurations of a static mixed-reactor. Each configuration was investigated in evaluating the effects of different shapes of obstacles on

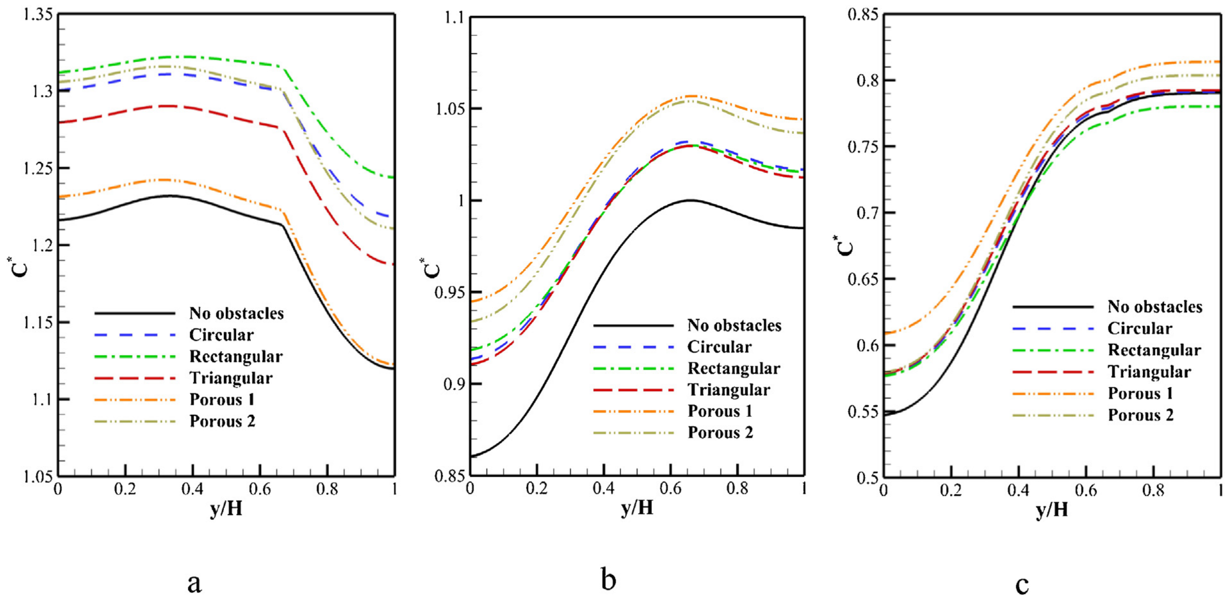

Fig. 4 – Normalized profile of nitrogen concentration at outlet for different geometries and inlet velocities in the left boundary: (a) $u = 0.5u_0$, (b) $u = 1.0u_0$, (c) $u = 1.5u_0$.

flow pattern and species mixing in a T-mixer–reactor. The effects of different parameters, including inlet velocity and concentration, on the denitrification rate have also been investigated numerically using LBM. For a better comparison among the results, all the concentration profiles are normalized using results obtained for clear channel $u = 1.0u_0$, $C_{1\text{-}1}$ = 100 mg-N/L and $C_{2\text{-}u}$= $C_{2\text{-}u0}$ = 100 mg-N/L.

5.1. *Effects of geometry*

Effects of different geometries on flow field, consequently mixing rate and the nitrate ion concentration field are shown in Fig. 3. Left inlet velocities of $0.5u_0$, $1.0u_0$, and $1.5u_0$ are applied, which are equal to inlet velocities of 3.55, 7.1, and 10.65 m/day, respectively. Reynolds number in the upper boundary was set up as half of the left inlet. Due to the 2:1 width ratio of the two inlets, this results in similar inlet velocity values. It is evident that increasing inlet velocity has apparent effects on the concentration fields. This is because mass transport depends on local velocity, as formulated in Eq. (4). All the obstacles and porous blocks within the reactor have greater effects on the flow and concentration fields. Figs. 3 & 4 show that inserting porous blocks and solid obstacles increased denitrification rates in all velocities, except for the rectangular obstacles at $u = 1.5u_0$. It can be seen that the normalized concentrations at a lowest velocity ($u = 0.5u_0$), are almost as twice as that at $u = 1.5u_0$ (Fig. 4). The rectangular obstacles have the greatest effects on the mixing and denitrification at low velocity ($u = 0.5u_0$). In contrast, at higher velocities, the porous blocks show higher nitrogen concentration profiles. This can be explained that porous obstacles can enhance the mixing due to increasing vortexes at the higher velocities (Fig. 3). This is because higher velocities would cause higher shear stresses, higher mixing rate and more nitrogen production (Fig. 4). This shows the importance of the static mixer design especially in lower velocities. However, at a higher inlet velocity, the differences decrease between different geometries. Furthermore, when $u = 0.5u_0$, the lowest concentration is seen at near to upper wall ($y/H = 1$). In contrast, increasing inlet velocity results in the lowest concentration at the bottom wall ($y/H = 0$). This is due to changes of flow pattern, as shown in Fig. 3. In addition, increasing inlet velocity causes higher nitrite penetration in the domain especially in the lower part of the domain (yellow color, as seen in Fig. 3), which leads to lower methanol concentration in that area of the domain. This indicates a lower denitrification rate in this region (Fig. 4).

5.2. *Effects of inlet concentration*

Figs. 5 and 6 show effects of obstacle geometries on denitrification at differing methanol inlet concentrations. It can be seen that the denitrification reaction is more complete with higher inlet concentrations of methanol as a main component of denitrification process (Fig. 6). The concentrations of nitrogen at the outlet increase as the inlet methanol concentrations increase from $C_{2-u} = 0.5C_{2-u0}$ to $C_{2-u} = 2.0C_{2-u0}$ (Fig. 6a–d). The denitrification efficacy decreases in the order of porous block 2, porous block 1, rectangular obstacles, circular obstacles, triangular obstacles, and no obstacles. It is interesting that the rectangular obstacle results in about a decrease of 3% in denitrification at $C_{2-u} = 0.5C_{2-u0}$, compared to clear channel. In contrast, an increase of 10% in denitrification is observed at $C_{2-u} = 2.0C_{2-u0}$. This is because that the flow field and mixing in the rectangular shapes were affected by the inlet velocity. However, porous blocks enhance denitrification in all inlet concentration as shown in Fig. 6. Physically, this can be explained because porous blocks can enhance mixing of species, leading to a higher denitrification rate. This reveals better performance if the porous block were implemented in the static mixer–reactor. Fig. 6b shows there were little differences among nitrogen concentrations at the outlet of the T-mixer–reactor with different obstacle configurations when $C_{2-u} = 1.0C_{2-u0}$ or $C_{2-u} = 0.5C_{2-u0}$. The concentration of nitrogen is lower at the bottom wall ($y/H = 0$) and there are smaller differences among the outlet concentration of different obstacles (Fig. 6a), compared to those at the higher inlet concentrations (Fig. 6c & d). This can be explained because lower concentrations of inlet carbon are not enough to achieve effective or complete denitrification reactions. However, it can be observed that the static mixer–reactor design can obviously

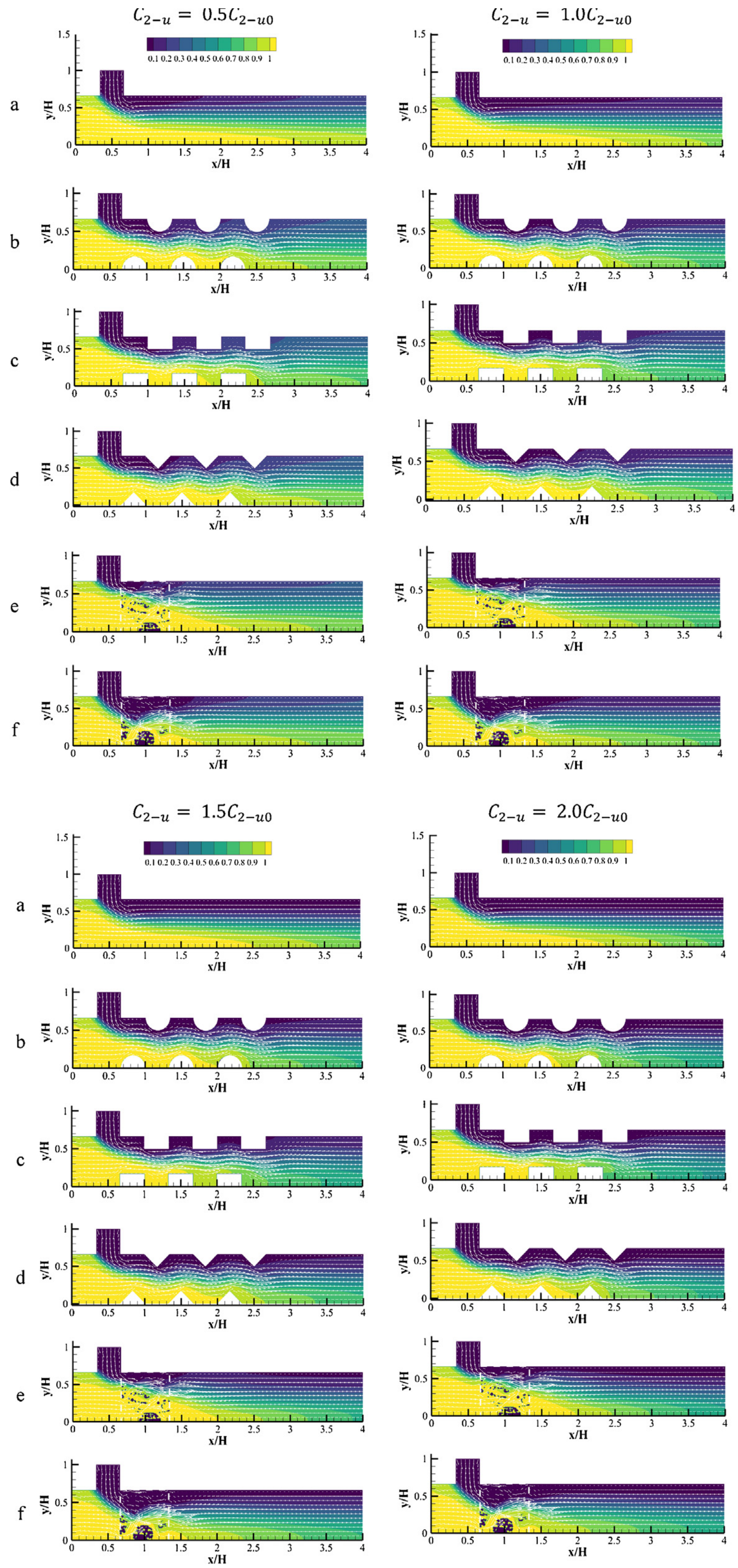

Fig. 5 – Normalized nitrate ion concentration in the domains: (a) no obstacles, (b) circular obstacles, (c) rectangular obstacles, (d) triangular obstacles, (e) porous media configuration 1, and (f) porous media configuration 2, at different inlet methanol concentrations of $C_{2-u} = 0.5C_{2-u0}$, $C_{2-u} = 1.0C_{2-u0}$, $C_{2-u} = 1.5C_{2-u0}$ and $C_{2-u} = 2.0C_{2-u0}$ (C_{2-u} is the inlet concentration of nitrate ions) (For interpretation of the references to colour in the text, the reader is referred to the web version of this article.).

increase the denitrification at higher inlet methanol concentrations. Compared to $C_{2-u} = 0.5C_{2-u0}$, all the configurations have higher performance with up to 8.6% enhancement than the clear channel when $C_{2-u} = 1.5C_{2-u0}$, and with up to a 10.3% improvement when $C_{2-u} = 2.0C_{2-u0}$.

6. Conclusion

In this study, a static mixer–reactor was designed for improving denitrification efficacy of wastewater treatment. A denitrification model was developed within lattice-Boltzmann

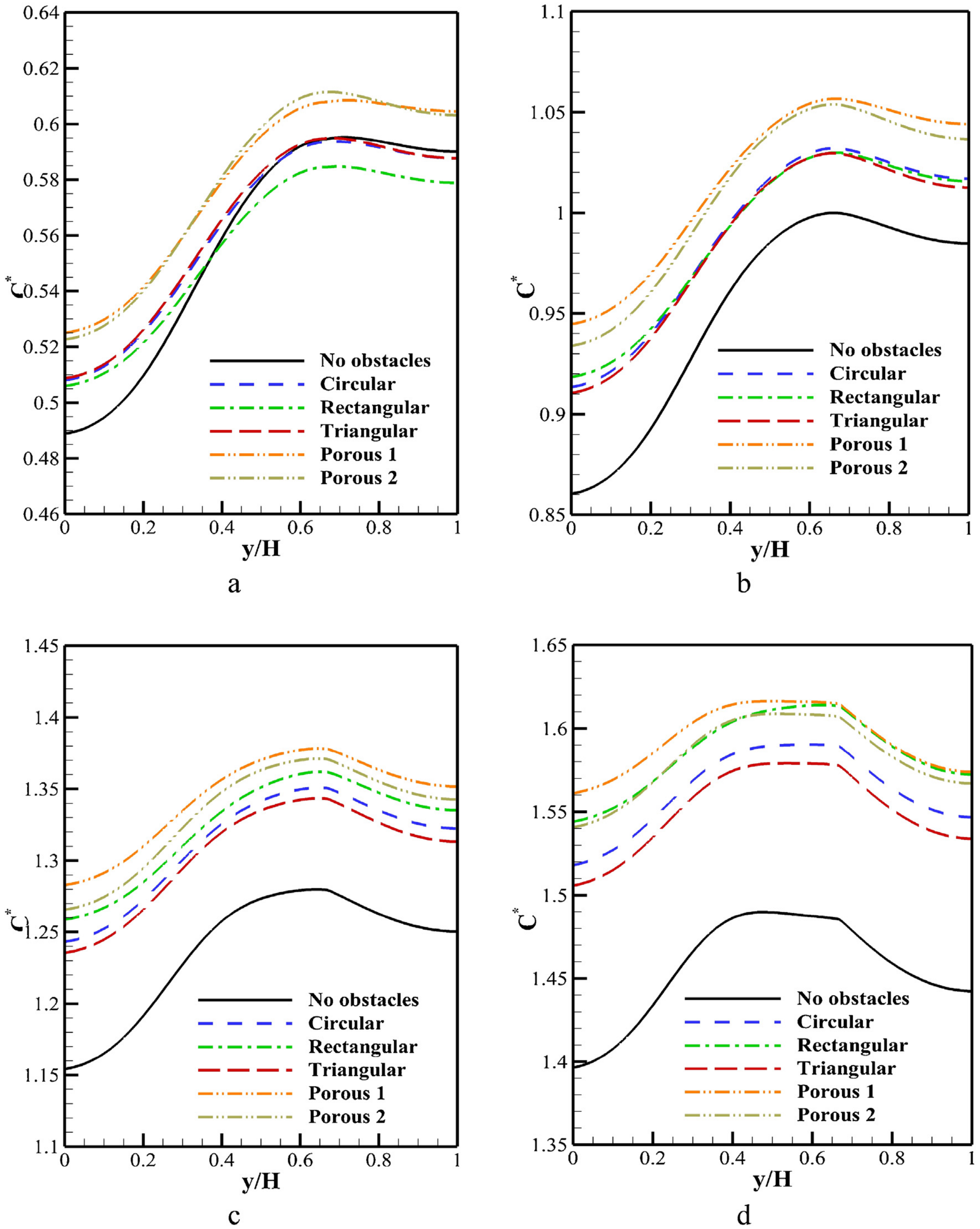

Fig. 6 – Normalized profile of nitrogen concentration at outlet for different geometries and inlet methanol concentrations: (a) $C_{2-u} = 0.5C_{2-u0}$ (b) $C_{2-u} = 1.0C_{2-u0}$, (c) $C_{2-u} = 1.5C_{2-u0}$, and (d) $C_{2-u} = 2.0C_{2-u0}$ in the upper inlet. C_{2-in} is the inlet concentration of methanol ions, and *y/H* represents the height of the coordinate at the outlet.

model (LBM). The model was calibrated and verified using the data available. LBM was then used to simulate the effects of six different obstacle configurations, different inlet velocities, and inlet methanol concentrations on denitrification in the T-mixer–reactor. The presence of obstacles increased the denitrification at every condition due to the increasing mixing but did have different performance under different operational conditions. It is found that the presence of obstacles had more impacts on the denitrification rate at a low inlet velocity than at a high inlet velocity. The increase in inlet velocity also changed the profile of the species in the reactor. The outlet nitrogen concentration showed an increase at the upper wall at a high inlet velocity compared to the lower wall at low inlet velocity. Thus, denitrification was more complete at lower inlet velocities as the nitrate had longer retention time in the reactor and more nitrates could be removed than that at higher inlet velocities. As a higher carbon source in denitrification processes, the methanol is often utilized for nitrate removal from the wastewater. The results show that when the inlet methanol concentration ($C_{2-u} = 0.5C_{2-u0}$) was low, there was a little effect of solid obstacles on the output concentration of nitrogen. However, the porous blocks did enhanced the denitrification at $C_{2-u} = 0.5C_{2-u0}$. All the obstacles and porous blocks significantly increased the denitrifications at higher concentrations ($C_{2-u} = 2.0C_{2-u0}$). Particularly the rectangular obstacle could lead to up to 10.3% improvement of the denitrification, which was comparable to the normalized nitrogen increases using the porous blocks. Therefore, the denitrification in the rectangular configurations could be affected considerably by the inlet methanol concentration. In contrast, at all the inlet methanol concentration, the notable performance improvements were observed in the porous block configurations. This implies that the porous configurations could be suitable for a wide range of operational conditions. This demonstrates that the proposed LBM model of denitrification processes can be capable of simulating complex processes of denitrification in wastewater treatment reactors and evaluating the performance of different reactor configurations.

Declaration of Competing Interest

The authors report no declarations of interest.

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